

Project Proposal: Real-Time Continuous American Sign Language (ASL) to Text Translator

1. Project Objective

This project is aimed at creating an effective, real-time translation system that will be able to translate Continuous American Sign Language (ASL) to textual content, written in English and specified fluent and grammatically correct. This system will take continuous video streams instead of discrete, already-segmented gestures, unlike traditional systems, which will have to contend with the complexities of co-articulation (smooth transitions between signs) and unique ASL syntax.

2. Proposed Architecture

The project will use a multi-stage pipeline composed of Computer Vision, Deep Learning, and Natural Language Processing to make use of real-time continuous translation:

- **Spatial Feature Extraction (MediaPipe):** The system will record an image in real time and relies on MediaPipe Holistic of Google to detect dense 3D spatial key points (hands, pose, and facial landmarks) per frame.
- **Temporal Sequence Modeling (Encoder-Decoder LSTM):** The obtained spatial features will be inputted in an Encoder-Decoder Long Short-Term Memory (LSTM) network. This will profile the time-relationships of the gestures with time.
- **Unsegmented Video (CTC Loss):** The model will be trained with Connectionist Temporal Classification (CTC) Loss to address the influence of the so-called boundary problem in which continuous signers do not interrupt between words. This enables the network to match unsegmented input frame to an output sequence of ASL glosses (raw word translations).
- **Syntax Translation (NLP/LLM Layer):** Since ASL has its own unique grammatical form (i.e. Store I go rather than I am going to the store), the raw ASL glosses will go through a Large Language Model (LLM) API to contextualize the signs and produce a fluent English sentence.

3. Technical Justification & Literature Review

The decision to use a Hybrid CNN-LSTM over a conventional 3D-CNN can be firmly supported by the latest literature, where the need to incorporate the concept of time through sign language dynamic modeling is brought to light. This was demonstrated by a 2024 study by MDPI on isolated video-based sign language recognition that indicated that a Hybrid CNN-LSTM significantly outperformed a pure 3D-CNN (I3D) with 84.65 percent, as compared to the pure CNN-based CNN with 65.89 percent. This sharp contrast proves that even conventional CNNs are unable to identify dependencies of sequence required in signs. In addition, a 2025 IEEE paper ("AI-Based Sign Language to Text Translator Using Computer Vision and Deep Learning") also used a similar spatial-temporal hybrid architecture, namely, sending MediaPipe-processed data to an LSTM, and reported an average accuracy of 92% in gesture classification, which fully confirms the feasibility and high performance of the suggested method.

4. Implementation Plan

The designing of this system shall be carried out in four major stages. First, the data preprocessing step will aim at deriving and normalizing spatial key points of the continuous ASL datasets (WLASL or MS-ASL) with the help of MediaPipe. Then, in model training, an Encoder-Decoder LSTM will be built and trained with CTC Loss to be able to process these unsegmented sign sequences. After the core recognition model has been running, the NLP integration step will bridge the raw gloss output of the model to an LLM to produce fluent and structured English sentences. Lastly, real-time optimization occurs through the sliding-window implementation to control video buffering that will ensure the entire pipeline achieves the requirement of latency, which is stringent to live translation at a minimum of 30 frames per second.

5. Expected Deliverables

- A trained, optimized Hybrid CNN-LSTM model capable of predicting continuous ASL glosses.
- A functional, real-time user interface that captures webcam input and displays translated, grammatically correct English sentences with minimal latency.
- A comprehensive final report detailing the model's accuracy, latency metrics, and environmental robustness.